

Research Methodology: Class-7-8



Class-7: Attitude Measurement and Scaling
Class-8: Questionnaire Designing

Recap from Class 5-6 and Connection to class 7-8

2

Step-3: Data Collection Methods

Learn during class 5-6

Planning stage-2 (Planning how)

End of the class we learn Methods and tools of data collection, Project techniques

Research Journey

Step-4: Continue with Attitude Measurement and Scaling, and Questionnaire Designing

Learn during class 7-8

Planning stage-2 (Planning how)

End of the class we will learn measurement, scaling and questionnaire designing at the planning stage of research

Learning Objective for Class-7

By the end of the class, the students must be able to understand and answer these questions?

- What is measurement in research?
- What are the types of measurement scales are there?
- Define Attitude and its three components.
- What are the various types of scales?
- Define measurement error and explain the criteria for good measurement.

Measurement in Research

4

The term measurement means assigning numbers or some other symbols to the characteristics of certain objects

Why do we need measurement?

- We need to measure the concepts in our hypotheses.
- When we measure those concepts, they become variables.
- Measurement is important for collecting data.
- So, measurement comes before data collection.

What is measurement?

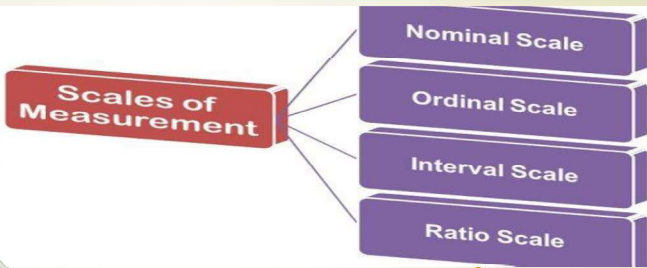
- **It is assigning numerical values or numerals to concepts according to some rule.**
- Measurement is the process of observing and recording the observations that are collected as part of a research effort.

Types of Measurement Scales

5

In research methodology, a scale is a tool to measure or represent the characteristics, attitudes, opinions, or behavior of people.

A scale helps us convert a person's response or characteristic into a form that can be measured and analyzed.



For example: If we want to know **how satisfied customers are**, we can ask them to rate their satisfaction from **1 to 5**.

Nominal Scale

6

- **Nominal scale** A scale in which the numbers used are only labels
- It is simply **a system of assigning number symbols to elements in order to label them**.
- **The number assigned in a nominal scale cannot be added, subtracted, multiplied, and divided.**
- **For example:** a population may be classified as males and females. Males can be assigned numeral as 1 and females as 2. Members of a labelled set can only be counted.
- Each number represents a particular object or response, **but the numbers have no value** apart from that: e.g. **drivers' licence, student ID, telephone number, credit card number**

7

Example of Nominal Scale

What is your gender?

- Male ✓
- Female

What is your hair colour?

- Brown
- Black ✓
- Gray
- Others

Type of Blood Group

- A
- B
- AB
- O

Remember:
Nominal = Name
It tells us "Which category?"

NOMINAL DATA

Nominal data divides variables into mutually exclusive, labeled categories.

Examples

Eye color
Blue Brown Green

Smartphone
iPhone Samsung Moto

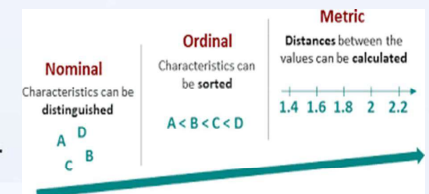
Transport
Bus Train Car

How is nominal data analyzed?

Descriptive statistics:
Frequency distribution and mode

Non-parametric statistical tests

Going to learn in later classes



Ordinal Scale

- Ordinal scale places elements in order. That means it tells whether an object has more or less of characteristics than some other objects.
- The **main characteristic of the ordinal scale is that the categories have a logical or ordered relationship**. This type of scale permits the measurement of degrees of difference, (i.e. 'more' or 'less') but not the specific number of differences (i.e. how much 'more' or 'less').
- For example:** ranking of students according to scores obtained by them
- For example:** A fast-food home delivery shop may wish to ask its customers: How would you rate the service of our staff? (1) Excellent • (2) Very Good • (3) Good • (4) Poor • (5) Worst •

Example of ordinal scale

How to you feel today?

- Very happy ✓
- Unhappy
- Ok
- Happy
- Very happy

Ordinal Scale involves the ranking of items along the continuum of the characteristic being scaled.

How satisfied are you with our service?

- Very unsatisfied ✓
- Somewhat unsatisfied
- Neutral
- Somewhat satisfied
- Very satisfied

Remember
Ordinal = Order
It tells us "Which comes first, second, third, etc.?"

ORDINAL DATA

Ordinal data classifies variables into categories which have a natural order or rank.

Examples

School grades



Education level



Seniority level

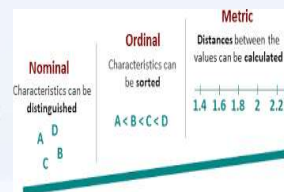


How is ordinal data analyzed?

Descriptive statistics:
Frequency distribution, mode, median, and range

Non-parametric statistical tests

Going to learn in later classes



Interval Scale

- Interval Scale is a scale in which **the numbers are used to rank attributes such that numerically equal distances on the scale represent equal distance in the characteristic being measured**.
- An interval scale **does not have a true zero**. e.g. A temperature of "zero" does not mean that there is no temperature...it is just an arbitrary zero point.
- An interval scale **contains all the information of an ordinal scale, but it also one allows to compare the difference/distance between attributes**.

Interval = Equal Difference + No True Zero

Interval Scale

13

Temperature
Suppose the temperature is:

Monday = 10°C

Tuesday = 20°C

Wednesday = 30°C

The difference is:

$20 - 10 = 10^\circ\text{C}$

and

$30 - 20 = 10^\circ\text{C}$

So, the intervals are equal.

But we cannot say that 20°C is twice as hot as 10°C, **because 0°C is not a true zero point.**

Remember:
Interval = Equal Intervals
It tells us "How much is the difference?"

INTERVAL DATA

Interval data is measured along a numerical scale that has equal intervals between adjacent values.

Examples

Temperature

90°

80°

70°

IQ score

40 100 160

Income ranges

\$19-29k \$30-39k \$40-49k

How is interval data analyzed?

Descriptive statistics: Frequency distribution; mode, median, and mean; range, standard deviation, and variance

Parametric statistical tests (e.g. t-test, linear regression)

We will learn in later classes

Ratio Scale

15

- The ratio scale has the **properties of an interval scale together with a fixed (absolute) zero point.**
- **The absolute zero point allows us to construct a meaningful ratio.**
- **Ratio scales permit the researcher to compare both differences in scores and relative magnitude of scores.**

16

Ratio Scale

Example: Weight

Suppose:

Person A = 20 kg

Person B = 40 kg

We can say:

Person B weighs twice as much as Person A.

Here, 0 kg represents the absence of weight, so zero is meaningful.

Remember:
Ratio = True Zero + Equal Intervals

RATIO DATA

Ratio data is measured along a numerical scale that has equal distances between adjacent values, and a true zero.

Examples

Weight in KG

...50 70 90...

Number of staff

...10 30 50...

Income in USD

...20k 40k 60k...

How is ratio data analyzed?

Descriptive statistics: Frequency distribution; mode, median, and mean; range, standard deviation, variance, and coefficient of variation

Parametric statistical tests (e.g. ANOVA, linear regression)

Will Learn it in later classes

Nominal

Which course are you studying?

BBA
BCA
MBA

Only categories → Nominal

Ordinal

How satisfied are you with your course?

Very dissatisfied
Dissatisfied
Neutral
Satisfied
Very satisfied

There is an order → Ordinal

Interval

What is the classroom temperature?

20°C
25°C
30°C

Equal differences → Interval

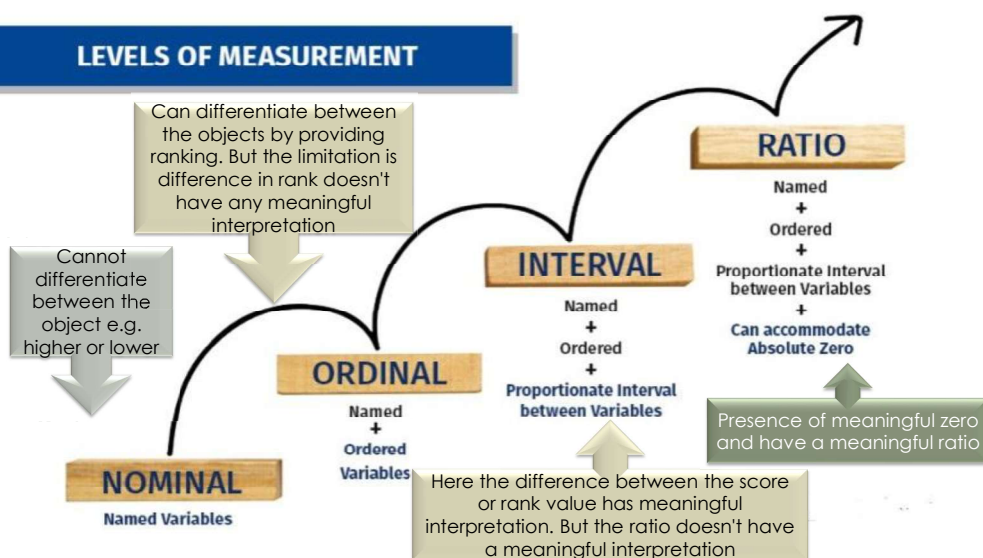
Ratio

How many hours do you study per day?

0 hours
2 hours
4 hours
6 hours

Equal intervals and a true zero → Ratio

LEVELS OF MEASUREMENT



MEASUREMENT SCALES

Nominal

Categories

Ordinal

Categories

Rank

Interval

Categories

Rank

Meaningful distance

Ratio

Categories

Rank

Meaningful distance

True Zero



What is Attitude?

21

Attitude means a person's positive or negative feeling, belief, or opinion toward a person, object, idea, or situation.

In simple words:

Attitude is the way a person thinks, feels, and tends to behave toward something.

Simple Example: Suppose a student thinks that **online classes are useful.**

- The student **believes** online classes are convenient.
- The student **likes** attending online classes.
- The student is **willing to attend** online classes regularly.

Together, these form the student's attitude toward online classes.

Three Components of Attitude: 1

22

Attitude has **three main components**:

1. Cognitive Component: "What I THINK"

The cognitive component refers to a person's beliefs, knowledge, thoughts, or opinions about something.

➤ **Example:** A student thinks:

"Online classes save travel time and provide flexibility."

This is the **cognitive component** because it represents what the student **believes or thinks**.

➤ **Easy way to remember: Cognitive = Thinking**

The cognitive component represents an individual's information and knowledge about an object.

Three Components of Attitude: 2

23

2. Affective Component – "What I FEEL"

The **affective component** refers to a person's **feelings or emotions** toward something.

➤ **Example:** The student says:

"I like online classes because they are comfortable."

This shows the student's **feeling** toward online classes.

➤ **Easy way to remember:**

Affective = Feeling

The affective component summarizes a person's overall feeling or emotions towards the objects

Three Components of Attitude: 3

24

3. Behavioural Component – "What I DO"

The behavioural component refers to a person's intention or tendency to behave in a particular way toward something.

Example: The student decides:

"I will attend my online classes regularly and participate in discussions."

This represents the **behavioural component** because it shows how the person is likely to **act**.

➤ **Easy way to remember:**
Behavioural = Doing/Acting

The behavioural component of an attitude reflects a predisposition to an action by reflecting the intention

Case-study-7.1: Exercise

25

- Let's take a very simple example: Attitude toward exercise.

Component	Meaning	Example
Cognitive	What I think	"Exercise keeps me healthy."
Affective	What I feel	"I enjoy exercising."
Behavioral	What I do/intend to do	"I exercise every morning."

Quick Formula to Remember

ATTITUDE = THINK + FEEL + ACT

Cognitive → Think

Affective → Feel

Behavioural → Act

Attitude is made up of what we THINK, what we FEEL, and how we are likely to ACT toward something.

Case-study-7.2: Examination

26

Imagine you are presenting this in class:

"Let us take the example of a student's attitude toward examinations.

- First, the **cognitive component** is what the student thinks, for example, 'Examinations are important for evaluating my knowledge.'
- Second, the **affective component** is what the student feels, for example, 'I feel confident and positive about examinations.'
- Third, the **behavioural component** is how the student acts, for example, 'I prepare regularly for my examinations.'
- Therefore, attitude has three components: **thinking, feeling, and behaviour.**"

27

Classification of Scales

Scaling

28

- Scaling is a method of measuring people's attitudes, opinions, preferences, perceptions, or behavior.**
- But what does that mean? In most scaling, the objects are text statements, usually statements of attitude or belief. The figure shows an example.
- To scale these statements, we have to assign numbers to them.**

the assignment...

...of objects...

...to numbers...

Are you willing to permit immigrants to live in your country?

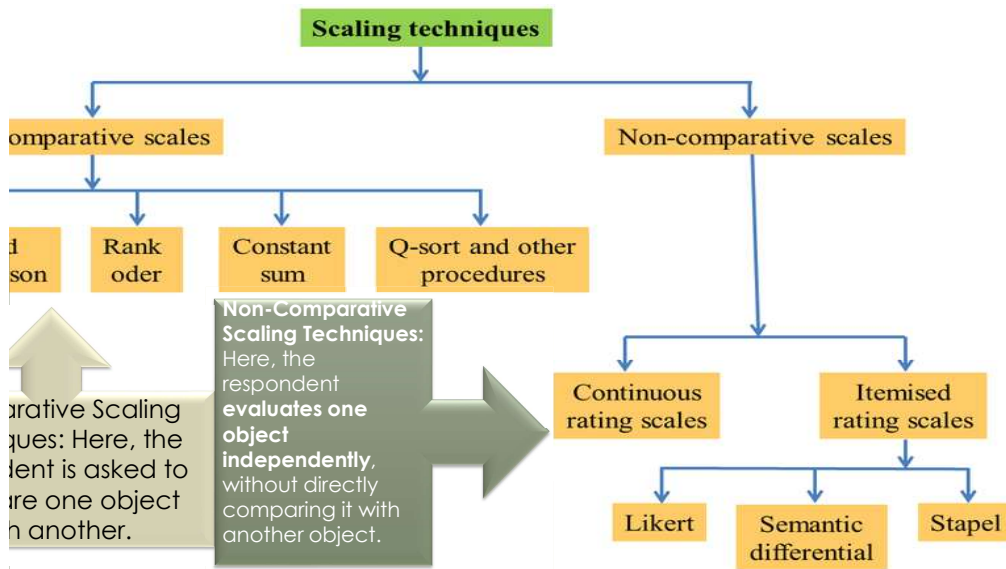
Are you willing to permit immigrants to live in your neighborhood?

Would you let your child marry an immigrant?

...according to a rule...



In research, **scaling** is the process of assigning numbers or symbols to the characteristics of people, objects, or responses according to specific rules.



1: Comparative Scaling Technique

30

Scaling Techniques

```

graph TD
    A[Scaling Techniques] --> B[Comparative]
    A --> C[Non-comparative]
    B --> D[Paired Comparison]
    B --> E[Rank Order]
    B --> F[Constant Sum]
    C --> G[Continuous rating or graphics rating]
    C --> H[Itemized rating]
    C --> I[Simple/multiple category scale]
    C --> J[Verbal frequency scale]
  
```

- In comparative scaling, **the respondent is asked to compare one brand or product against another.**
- For comparing two or more variables, **a comparative scale is used by the respondents.**
- Comparative scales involves **direct comparison of stimulus objects.**
- Example:** Respondent is asked to choose brand A toothpaste or brand B toothpaste

2: Non-Comparative Scaling Technique

31

Scaling Techniques

```

graph TD
    A[Scaling Techniques] --> B[Comparative]
    A --> C[Non-comparative]
    B --> D[Paired Comparison]
    B --> E[Rank Order]
    B --> F[Constant Sum]
    C --> G[Continuous rating or graphics rating]
    C --> H[Itemized rating]
    C --> I[Simple/multiple category scale]
    C --> J[Verbal frequency scale]
  
```

- In non-comparative scaling, **each object is scaled independently of the others.**
- In non-comparative scaling respondents need only **evaluate a single object.**
- Their **evaluation is independent of the other object** which the researcher is studying.
- Example:** The respondents are asked to give preference score on a 1 to 6 scale to brand A of the toothpaste. Here 1 is no-preferred at all,6 is highly preferred

Remember:
 Comparative = Compare
 Non-comparative = Evaluate independently

1 (a): Paired Comparison

32

- A paired comparison **symbolizes two variables from which the respondent needs to select one.** This technique is mainly used at the time of product testing, to facilitate the consumers with a comparative analysis of the two major products in the market.
- To compare more than two objects say comparing P, Q and R, one can first compare P with Q and then the superior one (i.e., one with a higher percentage) with R.

For example, A market survey was conducted to find out consumer's preference for the network service provider brands, A and B. The outcome of the survey was as follows:
 Brand 'A' = 57%
 Brand 'B' = 43%
 Thus, it is visible that the consumers prefer brand 'A', over brand 'B'.

33

1 (b): Rank order

- In rank order scaling the respondent needs to rank or arrange the given objects according to his or her preference.
- For example, A soap manufacturing company conducted a rank order scaling to find out the orderly preference of the consumers. It asked the respondents to rank the following brands in the sequence of their choice:

SOAP BRANDS	RANK
Brand V	4
Brand X	2
Brand Y	1
Brand Z	3

- The above scaling shows that soap 'Y' is the most preferred brand, followed by soap 'X', then soap 'Z' and the least preferred one is the soap 'V'.

1 (C): Constant Sum

34

- It is a scaling technique where a continual sum of units like dollars, points, chits, chips, etc. is given to the features, attributes and importance of a particular product or service by the respondents.
- For example, The respondents belonging to 3 different segments were asked to allocate 50 points to the following attributes of a cosmetic product 'P':

ATTRIBUTES	SEGMENT 1	SEGMENT 2	SEGMENT 3
Finish	11	8	9
Skin Friendly	11	12	12
Fragrance	7	11	8
Packaging	9	8	10
Price	12	11	11

- From the above constant sum scaling analysis, we can see that:
 - Segment 1 considers product 'P' due to its competitive price as a major factor.
 - But segment 2 and segment 3, prefers the product because it is skin-friendly.

35

2 (a): Non-Comparative Scales: Continuous Rating Scales or Graphic Rating

- It is a graphical rating scale where the respondents are free to place the object at a position of their choice. It is done by selecting and marking a point along the vertical or horizontal line which ranges between two extreme criteria.
- For example, A mattress manufacturing company used a continuous rating scale to find out the level of customer satisfaction for its new comfy bedding. The response can be taken in the following different ways (stated as versions here):



- The above diagram shows a non-comparative analysis of one particular product, i.e. comfy bedding. Thus, making it very clear that the customers are quite satisfied with the product and its features.

36

2 (b): Non-Comparative Scales: Itemized Rating Scale

- Itemized scale is another essential technique under the non-comparative scales. It emphasizes on choosing a particular category among the various given categories by the respondents. Each class is briefly defined by the researchers to facilitate such selection.
- The three most commonly used itemized rating scales are as follows:

Stapel Scale:

Semantic Differential Scale:

Likert Scale:

2(b-1) Likert Scale:

37

- 1 – Strongly Disagree
- 2 – Disagree
- 3 – Neither Agree Nor Disagree
- 4 – Agree
- 5 – Strongly Agree

- In the Likert scale, **the researcher provides some statements and ask the respondents to mark their level of agreement or disagreement over these statements by selecting any one of the options from the five given alternatives.**
- **For example,** A shoes manufacturing company adopted the Likert scale technique for its new sports shoe range named Z sports shoes. The purpose is to know the agreement or disagreement of the respondents. For this, the researcher asked the respondents to circle a number representing the most suitable answer according to them, in the following representation:

STATEMENT	STRONGLY DISAGREE	DISAGREE	NEITHER AGREE NOR DISAGREE	AGREE	STRONGLY AGREE
Z sports shoes are very light weight	1	2	3	4	5
Z sports shoes are extremely comfortable	1	2	3	4	5
Z sports shoes look too trendy	1	2	3	4	5
I will definitely recommend Z sports shoes to friends, family and colleagues	1	2	3	4	5

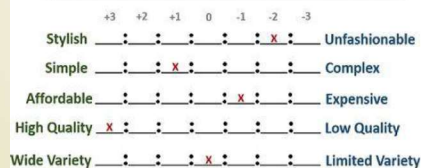
The above illustration will help the company to understand what the customers think about its products. Also, whether there is any need for improvement or not.

2(b-2) Semantic Differential Scale:

39

- **A bi-polar seven-point non-comparative rating scale is where the respondent can mark on any of the seven points for each given attribute of the object as per personal choice. Thus, depicting the respondent's attitude or perception towards the object.**
- **For example,** A well-known brand for watches, carried out semantic differential scaling to understand the customer's attitude towards its product. The pictorial representation of this technique is as follows:

Semantic Differential Scale

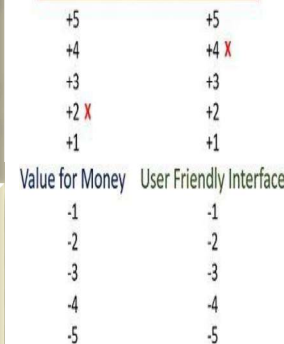


From the above diagram, we can analyze that the customer finds the product of superior quality; however, the brand needs to focus more on the styling of its watches.

2(b-3) Stapel Scale:

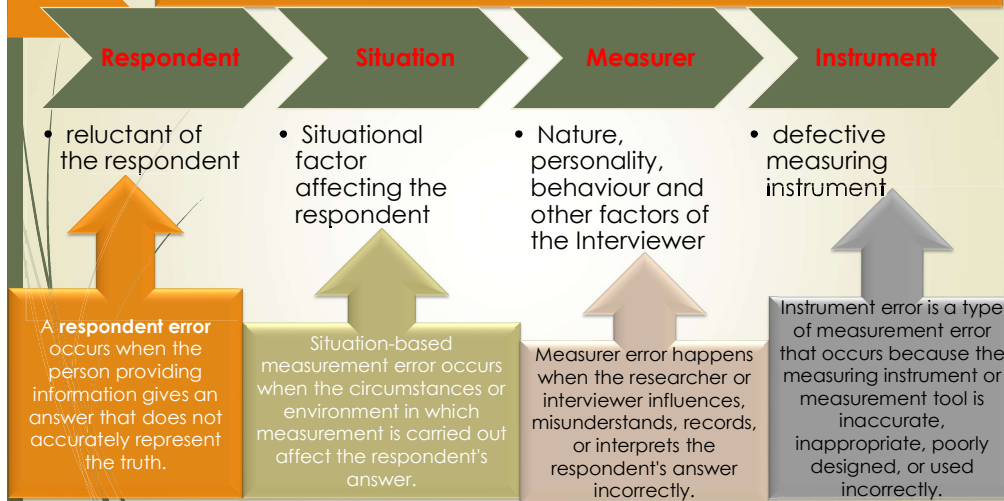
40

Stapel Scale

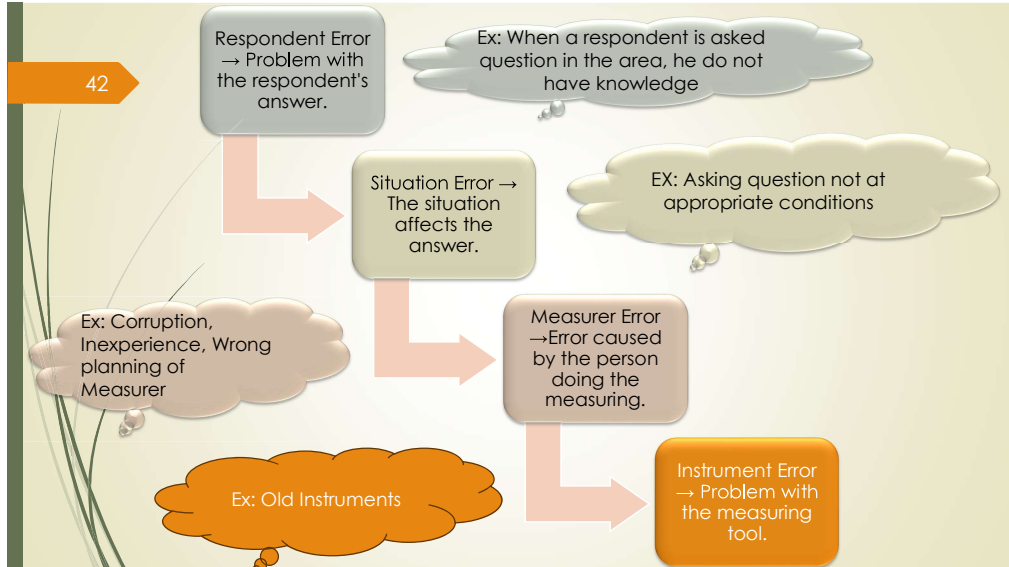


- A Stapel scale is that itemized rating scale which measures the response, perception or attitude of the respondents for a particular object **through a unipolar rating. The range of a Stapel scale is between -5 to +5 eliminating 0, thus confining to 10 units.**
- **For example,** A tours and travel company asked the respondent to rank their holiday package in terms of value for money and user-friendly interface as follows:
- With the help of the above scale, we can say that the company needs to improve its package in terms of value for money. However, the decisive point is that the interface is quite user-friendly for the customers.

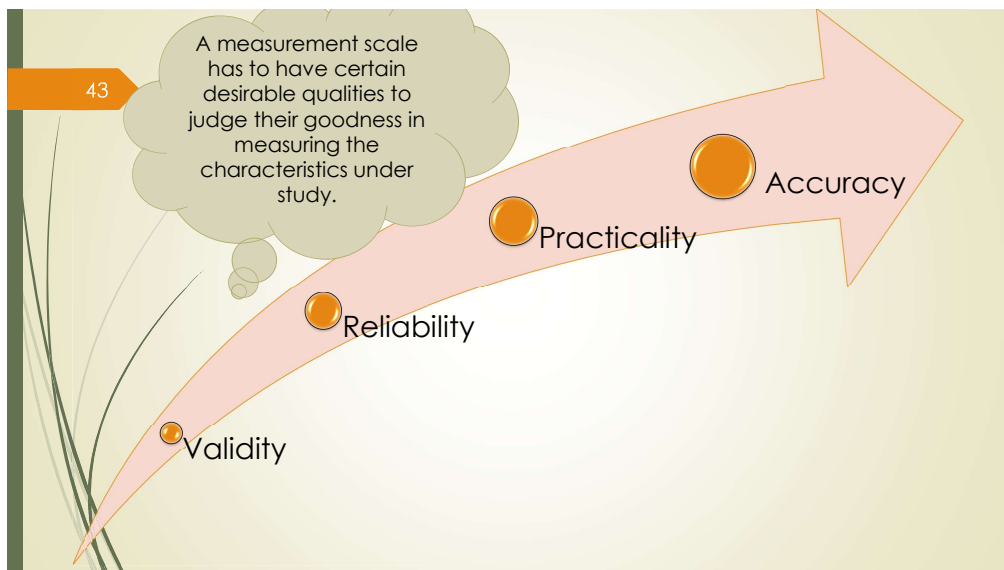
Sources of Error in Measurement



42



43



1: Validity

44

- Validity: Validity refers to the extent to which a research instrument, test, or measurement tool actually measures what it is intended to measure.
- In simple words: **Validity means the accuracy or correctness of a measurement.**
- Example:** If a researcher develops a questionnaire to measure students' academic motivation, the questionnaire is valid if it actually measures academic motivation and not intelligence, anxiety, or some other factor.
- Simple formula to remember: **Validity = Are we measuring the right thing?**
- Three types of validity**
 - Content validity.**
 - Criterion-related validity.**
 - Construct validity.**

45

1(a): Content validity

- **Content validity refers to the extent to which a research instrument or test adequately covers all the important areas or aspects of the subject or concept it is intended to measure.**
- **Example:** A mathematics test has good content validity if it includes questions covering algebra, geometry, statistics, probability, etc., according to their importance in the syllabus.
- In simple words: **Does the test cover the whole content properly?**

46

1(b): Criterion-related validity

- **Criterion-related validity refers to the extent to which the results of a test or research instrument are related to or agree with an established external criterion or standard.**
- **Example:** If a new entrance examination predicts students' actual academic performance in college, the examination has good criterion-related validity.
- In simple words: **Does the test agree with or predict a recognized external standard?**

47

1(C): Construct validity

- **Construct validity refers to the extent to which a research instrument actually measures the theoretical concept or construct that it is intended to measure.**
- **Example:** If a questionnaire is designed to measure anxiety, construct validity means demonstrating that it actually measures anxiety, rather than stress, depression, or some other concept.
- In simple words: **Does the instrument really measure the concept it claims to measure?**

48

2: Reliability

- A measuring instrument is **reliable if it provides consistent results.**
- Reliable measuring instrument does contribute to validity, but a reliable instrument need not be a valid instrument. **For instance, a scale that consistently overweighs objects by five kgs., is a reliable scale, but it does not give a valid measure of weight.**
- Two aspects of reliability **1.stability 2. equivalence**

49

Reliability: 2(a) Stability aspects

- The stability aspect is **concerned with securing consistent results with repeated measurements** of the same person and with the same instrument.
- We usually determine **the degree of stability by comparing the results of repeated measurements.**

50

Reliability: 2(b) Equivalence aspects

- The equivalence aspect **considers how much error may get introduced by different investigators or different samples of the items being studied.**
- A good way to test for the equivalence of measurements by two investigators is **to compare their observations of the same events.**

Validity = Are we measuring the right thing?
Reliability = Are we getting consistent results?

RELIABILITY



Reliable
Not Valid



Low Reliability
Low Validity



Not Reliable
Not Valid



Reliable
Valid

52

3: Practicality

- The practicality characteristic of a measuring instrument can be judged in terms of **economy, convenience and interpretability.**
- From the operational point of view, **the measuring instrument ought to be practical i.e., it should be economical, convenient and interpretable.**
 - **1: Economy** consideration suggests that some trade-off is needed between the ideal research project and that which **the budget can afford.**
 - **2: Convenience** test suggests that the measuring instrument should be easy to administer. For this purpose, one should give due attention to the **proper layout of the measuring instrument.**
 - **3: Interpretability** consideration is especially important when persons other than the designers of the test are to **interpret the results.**

53

4: Accuracy

- The characteristics of accuracy of a measurement scale **means it should be a true representative of the observation of underlying characteristic.**
- **For example**, measuring with an 'inch' scale will provide accurate value only up to one-eighth of an inch, while measuring with 'cm' scale will provide more accurate value.

54

Case-study-7.3: Teacher Setting a Question Paper

Suppose a **teacher prepares a question paper for a Class 10 Mathematics examination.**

- **Content validity:** The question paper must covers all important topics in the syllabus, such as **Algebra, Geometry, Trigonometry, and Statistics.**
- **Criterion-related validity:** Students' marks in the new question paper are similar to their performance in a **standardized/previous mathematics examination.** This need to be checked.
- **Construct validity:** The question paper actually measures **students' mathematical knowledge and problem-solving ability**, not mainly their memory or language ability.
- **Easy memory:**

Content → Covers the syllabus

Criterion → Matches another standard

Construct → Measures the intended ability

55

Case-study-7.4: The Teacher's Question Paper

A teacher prepares a Mathematics question paper for her students.

After the examination, she checks the paper in two ways:

- **Validity — "Am I testing the right thing?"**

The teacher wants to test **Mathematical ability.**

If the questions actually test mathematics, calculation, reasoning, problem-solving, etc., the paper has **validity.**

Validity = Are we measuring the right thing?

- **Reliability — "Am I getting consistent results?"**

The teacher gives a similar test to the same students under similar conditions. Their performance is **consistent and nearly similar.**

The test has **reliability.**

Reliability = Are we getting consistent results?

- **One-line story to remember:** Validity means the question paper tests what it is supposed to test; reliability means it gives consistent results.

Easy memory:
Validity = Accuracy
Reliability = Consistency

56

Class-8: Questionnaire Designing



Learning Objective for Class-8

57

By the end of the class, the students must be able to understand and answer these questions?



- ▶ What are the primary criteria for questionnaire designing?
- ▶ Mention about the questionnaire design procedure.
- ▶ What are the various types of questions that can be included in a questionnaire?
- ▶ Write a short note on the physical characteristics of a questionnaire

Criteria for Questionnaire Designing

58

1. Convert research objectives into specific questions

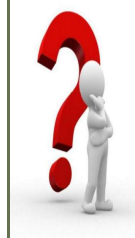
The clearly stated research objectives should be converted into questions that can obtain the required information from respondents. The questions should be framed in such a way that respondents can provide meaningful and useful answers.

2. Engage the respondent and encourage meaningful responses

The questionnaire should be designed to maintain the respondent's interest and motivate them to answer the questions properly and completely.

3. Use simple and self-explanatory language

Questions should be expressed in **simple, clear and understandable language**. They should be self-explanatory, particularly when the questionnaire is self-administered.



Case-study 8.1: Student satisfaction

59

Imagine a researcher wants to study **student satisfaction with college facilities**.

He asks:

"Are you satisfied with the excellent teaching, infrastructure and placement facilities provided by our college?"

This questionnaire has three problems.



Problem 1: Research objective is not properly converted:

60

If the objective is to measure satisfaction with different facilities, one question should not try to measure everything.

Better:

- ▶ How satisfied are you with the **teaching quality**?
- ▶ How satisfied are you with the **college infrastructure**?
- ▶ How satisfied are you with the **placement facilities**?



61

Problem 2: It may not encourage an honest response:

- Calling the facilities "excellent" as mentioned in the question asked may push students toward a positive answer.
- Survey-design guidance recommends avoiding leading questions because they can bias responses.

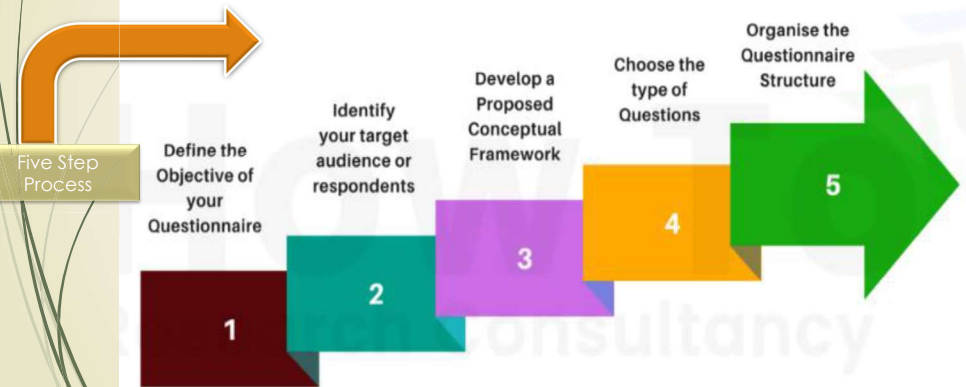


Problem 3: The question is not simple:

A student may be satisfied with teaching but dissatisfied with infrastructure. Putting several concepts into one question creates a **complex question**, making the response difficult to interpret.

62

Questionnaire design procedure: How to prepare questionnaire for research



63

Types of question response options

Open-ended

Closed-ended

Type of Questions

Dichotomous

Multiple responses

Scales

64

Determine the type of questions

- Broadly, questions can be classified into **structured and unstructured questions**.

1. Structured Questions

- Definition:** A structured question provides the respondent with a pre-determined set of alternative answers, from which the respondent has to choose.

- Example:** Do you like working from home?

- ☐ Yes
- ☐ No

2. Unstructured Questions

Definition: An unstructured question gives the respondent complete freedom to answer in his or her own words.

Example:

What problems do you face while working from home?

The respondent can write any answer, such as:

"I find it difficult to separate my personal and professional life."

Unstructured questions are useful when the researcher wants to explore opinions, feelings, experiences and motives in depth.

Types of Questions Used in a Questionnaire

65

1. Open-ended Questions

Definition: An open-ended question allows the respondent to answer freely in his or her own words, without providing a fixed list of alternatives.

Example: What are the major challenges you face while working from home?

Possible answers could be:

- Poor internet connection
- Lack of concentration
- Work-life imbalance
- Lack of interaction with colleagues

Why use it?

Open-ended questions are useful when the researcher wants **detailed and in-depth information** and does not want to restrict the respondent's answer.

Easy way to remember

Open-ended = "You tell me in your own words."

Continue...

66

2. Closed-ended Questions

Definition: A closed-ended question provides the respondent with a set of alternative answers in advance, and the respondent has to choose one of them.

Example: Do you like working from home?

- ☐ Yes
- ☐ No

Why use it?

They are:

- Easy for respondents to answer
- Easy to code
- Easy to compare
- Easy to analyse statistically

Easy way to remember

Closed-ended = "Choose from the answers I give you."



Continue....

67

3. Dichotomous Questions

Definition: A dichotomous question is a type of closed-ended question that provides only two alternative answers.

Common pairs include:

- Yes / No
- True / False
- Agree / Disagree
- Male / Female (where appropriate to the study's design)

Example: Does working from home save travelling time?

- ☐ True
- ☐ False

Another example: **Do you use online banking?**

- ☐ Yes
- ☐ No

Why use it?: Dichotomous questions are **simple to answer** and produce **clear-cut responses**.

Easy way to remember

Dichotomous = "Only two choices."



Continue...

68

4. Multiple-choice Questions

Definition: A multiple-choice question provides more than two alternative answers, and the respondent selects the most appropriate answer.

Example: How often do you work from home?

- ☐ Every day
- ☐ 2-3 days a week
- ☐ Once a week
- ☐ Rarely
- ☐ Never

Why use it?: They are useful when there are **several possible alternatives** from which respondents can choose.

Easy way to remember

Multiple-choice = "Many options, choose one."

Continue...

69

5. Rating Scale Questions

Definition: A rating scale question asks the respondent to indicate their degree of agreement, disagreement, satisfaction, importance, preference, etc., on a continuum or scale.

Example: Rate your satisfaction with working from home.

Rating	Meaning
1	Very dissatisfied
2	Dissatisfied
3	Neutral
4	Satisfied
5	Very satisfied



Physical characteristics of the questionnaire

70

The physical characteristics of a questionnaire include its overall appearance, layout, paper quality, and formatting that directly influence respondent cooperation and data accuracy.

Key Physical Characteristics

- **Appearance and Attractiveness:** An attractive and neat layout encourages people to complete the form instead of ignoring it.
- **Paper Quality and Colour:** Using good-quality paper and pleasant, non-distracting colors makes the questionnaire look professional and readable.
- **Size and Format:** The physical dimensions and page size should be manageable, clear, and convenient to handle or mail.
- **Spacing and Layout:** Proper margins, adequate blank space for written answers, and clear separation between questions prevent confusion and mistakes.
- **Typography and Graphics:** Clear fonts, legible text sizes, and simple visual elements help guide the respondent smoothly through the sections.
- **Design Based on Administration:** The format changes depending on who records the answers; self-administered forms need very explicit instructions, while interviewer-filled forms prioritize a smooth flow for the researcher.

71



Thanks